Project 1: Data Analysis Project

- **Vision:** *Programming is more than writing code*. The ultimate goal of the projects in this course is that you learn to structure a programming project from start to finish. Your answers should of course be correct, but your code should also be well-structured and documented.
- **Objectives:** In your data analysis project, you should show that you can
  1. Download online data using an API
  2. Simulate data to analyze
  3. Apply data cleaning and data structuring methods
  4. Apply data analysis methods
  5. Present results in text form and in easy-to-understand figures
  6. Teach yourself new functionality in a package you already know
  7. Structure a code project
  8. Document code
- **Structure:** Your data analysis project should consist of
  1. A single self-contained notebook (.ipynb) presenting the analysis
  2. Documented Python module files (.py) (if any)

  Example of structure: See this repository and

  `ProgEcon-lectures/Exams/Structure/`
- **Hand-in:** Upload in Absalon
- **Deadline:** See Calendar.
- **Feedback:** Your TA will provide feedback on your project.
- **Exam:** Your data analysis project will be a part of your exam portfolio.
  You can incorporate feedback before handing in the final version.
  This first version does not need to be perfect!

1 Inequality in Denmark

We use two tables from Statistics Denmark, both based on **equivalised disposable income**, i.e. income after taxes and transfers, adjusted for household size.

1.1 The Gini coefficient and the top 10 percent share

Using table `IFOR41` from Statistics Denmark, produce a figure with the Gini coefficient for all of Denmark over time.

Using table `IFOR32`, add the share of total income going to the richest 10 percent.

Describe how inequality in Denmark has developed since 1987. Do the two measures tell the same story? Compute the correlation between them over time.

Hint I: In `IFOR41` the variable `ULLIG` selects the inequality measure. The Gini coefficient is `ULLIG = 70`.

Hint II: `IFOR32` reports the *average* income within each decile group. Since every decile group contains the same number of people, the income share of the top decile is

$$\text{top 10 pct. share} = \frac{\bar{y}_{10}}{\sum_{d=1}^{10} \bar{y}_d}$$

Hint III: `pd.merge(...,validate='1:1')` is a cheap way to check that you have not accidentally duplicated rows.

1.2 Prediction

Fit a **linear** and a **quadratic** trend to the Gini coefficient, and use them to predict inequality in 2030 and 2040.

Fitting a polynomial trend of degree p by least squares means choosing the coefficients $\beta = (\beta_0, \ldots, \beta_p)$ that solve

$$\min_{\beta} S(\beta) = \sum_i \left(y_i - \sum_{k=0}^{p} \beta_k t_i^k \right)^2$$

1. **Solve this minimization problem yourself** with `scipy.optimize.minimize`.
2. **Check your answer** against `np.polyfit`. Do you get the same coefficients?
3. Use your estimates to predict the Gini coefficient in 2030 and 2040, and plot the result.

How well do the two specifications fit the historical data? How much do their predictions differ? Would you trust either of them, and why?

Hint I: Be careful with the *scaling* of the regressor. If you use the year itself, then t^2 is of the order $4 \cdot 10^6$ while $t^0 = 1$, and the optimizer will fail. Measuring time as *years since 1987* solves the problem. It is worth running it both ways to see what happens.

Hint II: You can evaluate a polynomial with `np.polyval`.

1.3 Municipalities

Both tables are also available at the municipality level. Do the following:

1. Compute the correlation between the Gini coefficient and the top 10 percent share *across municipalities*. Is it similar to the correlation over time you found above?
2. Find the 10 most and the 10 least unequal municipalities in the most recent year, and illustrate them in a figure. Describe the pattern.
3. Find the municipalities with the largest and the smallest change in the Gini coefficient. Illustrate and comment.

Warning: Before you interpret any ranking, *plot or tabulate the underlying series for the municipalities at the top and the bottom of it*. The Gini coefficient of a small population is not robust to a single very large income.

1.4 Extension

Find some other information at the municipality level and relate it to the level of the Gini coefficient or to its change over time.

Note: This is intended as an open question, and we encourage you to be creative.

2 Simulating the income distribution

We now simulate an income distribution from a simple life-cycle model.

Implement the model below in a documented `.py` **module** and import it into your notebook.

2.1 The model

A cohort of N individuals is followed from age 18 until retirement at age 65.

Education. At age 18 everybody is uneducated and outside the labor market. Each individual i draws an education $e_i \in \{\text{short}, \text{medium}, \text{long}\}$ with probabilities p_e. The education takes S_e years, during which the individual receives a student grant y^{SU}.

Labor market. After finishing their education, the individual enters the labor market as unemployed. Employment then follows a persistent two-state Markov chain: an unemployed person finds a job with probability λ, and an employed person loses their job with probability σ.

Human capital. Individuals enter the labor market with education-specific human capital $h_{e,0}$. It then evolves as

$$h_{i,t+1} = \begin{cases} h_{i,t}\,(1+\Delta_{e_i})\,\psi_{i,t+1} & \text{if employed} \\ h_{i,t}\,(1-\delta)\,\psi_{i,t+1} & \text{if unemployed} \end{cases}$$

where Δ_e is an education-specific growth rate, δ is the depreciation rate while unemployed, and ψ is a mean-one lognormal shock, $\log\psi \sim \mathcal{N}(-\frac{1}{2}\sigma_\psi^2, \sigma_\psi^2)$. Human capital is unchanged while in education.

Hint: Set up a random number generator with `rng = np.random.default_rng(seed)` and draw the shock with `psi = rng.lognormal(-0.5*sigma_psi**2, sigma_psi, size=N)`. This is the same as `psi = np.exp(rng.normal(-0.5*sigma_psi**2, sigma_psi, size=N))` -- note that the two arguments are the mean and the standard deviation of the *underlying normal*, not of the lognormal itself. The $-\frac{1}{2}\sigma_\psi^2$ is what makes the mean exactly one: without it the mean would be $\exp(\sigma_\psi^2/2)$. Check it with `psi.mean()`.

Income.

$$y_{i,t} = \begin{cases} y^{SU} & \text{in education} \\ h_{i,t} & \text{if employed} \\ \rho y_i^{\text{last job}} & \text{if unemployed} \end{cases}$$

i.e. an unemployed person receives the replacement rate ρ of the income in the *previous* job. Use a floor $\underline{y}$ for those who have never been employed.

Parameters.

Parameter	Symbol	Value
Number of individuals	N	50,000
Education probabilities	p_e	(0.40, 0.35, 0.25)
Years of education	S_e	(1, 3, 5)
Initial human capital	$h_{e,0}$	(1.00, 1.20, 1.55)
Growth of human capital	Δ_e	(0.010, 0.020, 0.030)
Depreciation while unemployed	δ	0.06
Std. of shock	σ_ψ	0.10
Job-finding probability	λ	0.60
Job-separation probability	σ	0.05
Student grant	y^{SU}	0.45
Replacement rate	ρ	0.60
Benefit floor	$\underline{y}$	0.35

2.2 Simulate the income distribution

1. Simulate the model. **Check that your simulation is correct** before you use it: the education shares should match p_e, and the unemployment rate should settle at $\sigma/(\sigma+\lambda)$, which can be shown to be the steady-state value.
2. Plot the mean and selected percentiles of income over the life cycle.
3. Plot a histogram of the income distribution at ages 25, 35, 45 and 60, and describe how the shape of the distribution changes.

2.3 Compute the Gini coefficient

Write your own function that computes the Gini coefficient of a vector of incomes.

Hint: Test it on cases where you know the answer. The Gini coefficient of a uniform distribution on $[0,1]$ is exactly $1/3$, and of a lognormal distribution whose logarithm has standard deviation s it is $2\Phi(s/\sqrt{2})-1$.

1. Compute the Gini coefficient for the full simulated sample. Plot the Lorenz curve.
2. Compute it separately for each age. How does inequality within an age group compare with inequality when all ages are pooled?

2.4 What drives inequality?

Run a series of alternative simulations, where the components of the model are switched off one at a time: educational differences, shocks to human capital, depreciation of human capital while unemployed, and unemployment itself. Report the Gini coefficient for each simulation and compare it to the baseline. Which mechanism matters the most?

Hint: Report the Gini coefficient both for all ages pooled and within a single age group. They need not move in the same direction, and where they do not, you should explain why.

2.5 Extension: more risk

Extend the model with another source of risk of your own choosing, and analyze how it changes the income distribution and the Gini coefficient.

Note: This is intended as an open question, and we encourage you to be creative.